CALIGA Setup Guide

1. Before you start

Ask James H. for these first. You can't finish setup without them.

What	Why
Access to the GitHub repo `jhepburn-hue/Internal-Tool-CALIGA`	To download the code
A filled-in `.env` file, or its values	Database, Slack, Jira, PocketBase and Forge settings
`settings_default.ini` and `customer_profiles.json`	CALIGA needs them to generate INI files. They're company data: keep them off shared drives and out of Git.
Read access to the `wavelynx_apex_config` Google Cloud bucket	Downloading INIs and binaries
A CALIGA user account (James adds you on the Manage Users page)	Only listed users can sign in
Forge access in your browser	Building binaries uses your own Forge sign-in

You'll use a terminal throughout. On Mac, open **Terminal** (Applications > Utilities). On Windows, open **PowerShell** (Start menu, type PowerShell).

> CALIGA connects to the team's real database, Slack channel and Jira project. Anything you create, fail or flash shows up for everyone.

2. Install the tools

You need three programs: Python 3.14.4, Git, and the Google Cloud CLI.

Mac

1. **Python.** Download the macOS installer for Python 3.14.4 from python.org/downloads and run it. When it finishes, double-click **Install Certificates.command** in the Python 3.14 folder it opens.
2. **Git.** In Terminal, run `xcode-select --install` and accept the prompt. Skip this if `git --version` already prints a version.
3. **Google Cloud CLI.** Follow the macOS steps at cloud.google.com/sdk/docs/install.
4. Check all three:

```
python3 --version
git --version
gcloud --version
```

Windows

1. **Python.** Download the Windows installer (64-bit) for Python 3.14.4 from python.org/downloads. On the first screen, tick **Add python.exe to PATH**, then choose **Install Now**.
2. **Git.** Install Git for Windows from git-scm.com/download/win. The default options are fine.
3. **Google Cloud CLI.** Run the Windows installer from cloud.google.com/sdk/docs/install.
4. Close PowerShell and open a new window, so it picks up the new programs. Then check all three:

```
py --version
git --version
gcloud --version
```

If the Python check shows a version other than 3.14, use `py -3.14` wherever this guide says `py`.

3. Download CALIGA and install its packages

This puts CALIGA in a folder on your Desktop and installs its Python packages into a private environment (`.venv`) inside that folder. Every package version is fixed in `requirements.txt`, so everyone gets the same versions.

Mac

```
cd ~/Desktop
git clone https://github.com/jhepburn-hue/Internal-Tool-CALIGA.git
cd Internal-Tool-CALIGA
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
```

Windows

```
cd $HOME\Desktop
git clone https://github.com/jhepburn-hue/Internal-Tool-CALIGA.git
cd Internal-Tool-CALIGA
py -m venv .venv
.venv\Scripts\pip install -r requirements.txt
```

The first `git clone` asks you to sign in to GitHub. Use the account James gave repo access to.

The install takes a minute or two. It's done when it prints `Successfully installed ...` with no red `ERROR` lines.

To get updates later, run `git pull` in the CALIGA folder, then run the install line again.

4. Configure

The private INI files

Make a folder outside the CALIGA folder and put `settings_default.ini` and `customer_profiles.json` in it. For example:

- Mac: `/Users/<you>/Documents/caliga-private`
- Windows: `C:/Users/<you>/Documents/caliga-private`

Keep them out of the CALIGA folder, so they can never end up in Git.

The .env file

Put the `.env` file James sent you in the CALIGA folder, next to `app.py`. It must be named exactly `.env`, with no `.txt` on the end. On Windows, turn on **View > Show > File name extensions** in File Explorer to check.

Then open it in a text editor and set these yourself:

Setting	What to put
`SECRET_KEY`	A new random value. Generate one with `python3 -c "import secrets; print(secrets.token_hex(32))"` (Windows: `py -c ...`).
`AUTH_MODE`	`dev`
`INI_PRIVATE_DIR`	The private folder from above. On Windows use forward slashes: `C:/Users/<you>/Documents/caliga-private`.
`ACTIVE_IAP_COOKIE` and `ACTIVE_IAP_UID`	Your own Forge sign-in. See below.

Leave the other values as James set them. Never send your `.env` to anyone or commit it. It holds passwords and tokens.

Google Cloud sign-in

Run this once and sign in with your WaveLynx Google account:

```
gcloud auth application-default login
```

CALIGA uses this sign-in to read INIs and binaries from the bucket.

Forge cookie

CALIGA starts Forge builds with your browser's Forge sign-in. It expires, so you'll repeat this when builds start failing.

1. Open Forge in Chrome and sign in.
2. Press F12 (Mac: Cmd+Option+I) and open **Application > Cookies**, then the Forge site.
3. Copy the value of the cookie whose name starts with `__Host-GCP_IAP_AUTH_TOKEN` into `ACTIVE_IAP_COOKIE`.
4. Copy the value of the `GCP_IAP_UID` cookie into `ACTIVE_IAP_UID`.
5. Save `.env` and restart CALIGA.

5. Run CALIGA and sign in

Start it from the CALIGA folder:

```
# Mac
cd ~/Desktop/Internal-Tool-CALIGA
.venv/bin/python app.py
```

```
# Windows
cd $HOME\Desktop\Internal-Tool-CALIGA
.venv\Scripts\python app.py
```

It's running when the terminal shows `Running on http://127.0.0.1:5000`. Leave that window open: closing it stops CALIGA. To stop it on purpose, press Ctrl+C in that window.

1. Open http://127.0.0.1:5000 in Chrome or Edge.
2. Open the user menu at the top right and pick your name.
3. You'll see a yellow "Development login" banner on every page. That's expected for now.

CALIGA only runs on your own computer. Other people can't open your copy, and you can't open theirs.

6. Flashing a reader

Only needed if you'll flash readers. You need the QA or SET role.

- **Browser:** Chrome or Edge. Flashing uses Web Serial, which Safari and Firefox don't have.
- **Adapter:** a USB RS-485 adapter wired to the reader's OSDP lines, plugged into the same computer that runs the browser.
- **Drivers:** Mac needs none. On Windows, plug the adapter in and wait for Windows to finish setting it up. If it doesn't show as a COM port in Device Manager, install the adapter maker's driver (for FTDI cables, the VCP driver from ftdichip.com).
- **Other programs:** close anything else using the adapter, such as a serial terminal or the old flashing tool. Only one program can use the port at a time.

When you press **Flash Configuration**, the browser asks which port to use. Pick the RS-485 adapter (on Mac it's usually `cu.usbserial-...`, on Windows a `COM` port).

7. Troubleshooting

Most problems show a message in the browser or in the terminal running CALIGA. Find it here.

You see	Fix
`python3`, `py` or `git` is "not found" or "not recognized"	Close and reopen the terminal. On Windows, reinstall Python with **Add python.exe to PATH** ticked.
The package install fails with red `ERROR` lines	Check `python3 --version` (Windows: `py --version`) shows 3.14, delete the `.venv` folder, and repeat step 3.
`Port 5000 is in use by another program`	Another copy of CALIGA is already running: close that terminal window. On Mac, also turn off **AirPlay Receiver** in System Settings > General > AirDrop & Handoff.
CALIGA stops at startup mentioning `SECRET_KEY` or `DATABASE_URL`	`.env` is missing, misnamed (`.env.txt`), or missing that value.
A change to `.env` has no effect	Stop CALIGA with Ctrl+C and start it again. It reads `.env` only at startup.
Your name isn't in the user menu, or you're told you can't sign in	Ask James to add or reactivate you on the Manage Users page.
"Can't generate INIs: set INI_PRIVATE_DIR" or "settings_default.ini isn't in INI_PRIVATE_DIR"	Fix the `INI_PRIVATE_DIR` path in `.env` (Windows: forward slashes), check both files are in that folder, and restart.
Errors about Google credentials, or downloads failing with 403	Run `gcloud auth application-default login` again. If it still fails, ask James for bucket access.
"Forge rejected the IAP cookie"	Your Forge cookie expired. Copy a fresh one (step 4) and restart.
Get Tokens or Sync fails with a PocketBase 403	The PocketBase token has expired. Ask James for a new one.
"Flashing needs Chrome or Edge"	Use Chrome or Edge on the computer the adapter is plugged into.

You see	Fix
"Could not open the adapter"	Close any other program using the adapter, unplug it, plug it back in, and try again.
The adapter isn't in the port list	Check the cable. On Windows, check Device Manager for a COM port and install the driver (step 6).